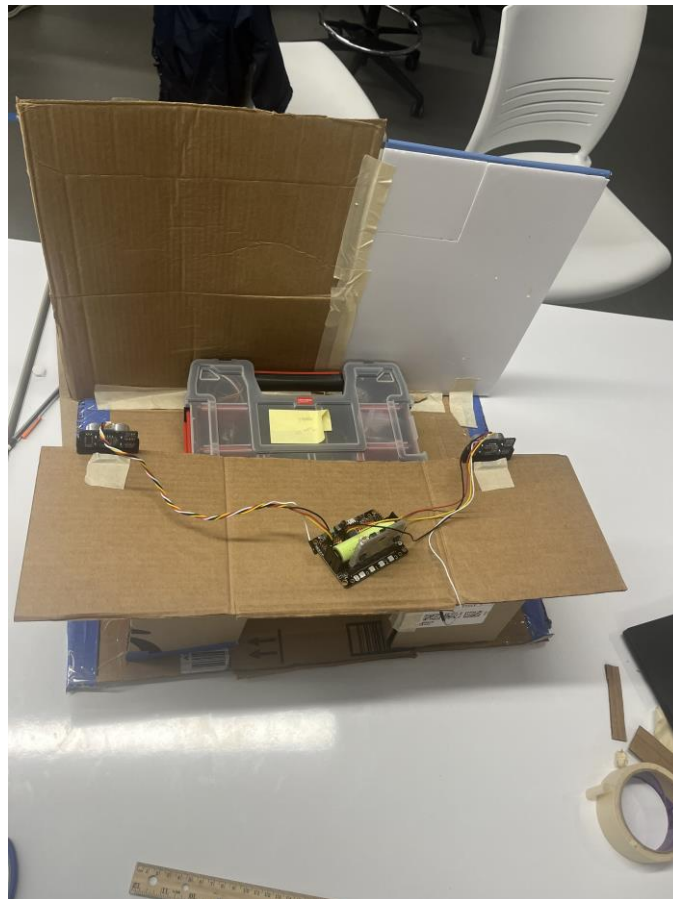


PROJECT 3 (StudyTrack) FINAL REPORT

Transforming Ideas to Innovation I&II

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Team 22

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I. COVER LETTER (DAVID)

The main problem the team decided to solve was that students waste their time looking for study spaces. Especially when final exams are close, students can have a hard time looking for a spot to study. From libraries to coffee shops, everywhere is packed with people making it too easy to waste time going from place to place looking for an opening. Simply put, we wanted to fix this by providing students with the information they need – locations of open places to work.

The solution we came up with was to have sensors that track people going in and out of the room and an app or display of some kind that will show how many people are in the room. The core of the team's design consisted of two ultrasonic sensors connected to a microbit for processing data. The ultrasonic sensors collected data and sent it to the microbit which kept track of the room to update and the count of people in the room. The algorithm chosen to count the number of people in the room was extremely simple. Since the ultimate challenge was to keep track of the direction people are moving, the final product used two sensors and compared the distances that both of them were reading. Since both sensors should read the same default number when pointed at the same wall, if one is lower than the other, then that is the side a person is entering from. The sensors then pause for a moment to avoid triggering the other sensor as the person passes through the hallway and continue reading after a reasonable amount of time to walk through has passed.

The design worked well, although there were some clear ways to improve the design. The product successfully met the requirements of being easy to use and cheap with it being made of the materials provided to us and focusing more on the software side of things. It was accurate with a caveat of it being less accurate when something moved through too quickly or extremely slowly. Additionally it could not read multiple people entering at exactly the same time. To get exact numbers of how accurate it would be in a

real world situation it would need to be placed at a popular spot to fully understand how often those scenarios occur and would be tracked incorrectly.

II. EXECUTIVE SUMMARY (DAVID)

The problem we solved was that students have no way of knowing how full a certain room is, and they can waste hours looking for a place to study during the most important time of the year. Popular study spots fill up extremely quickly during dead week and finals week around Purdue, and going from library to library looking for a quiet spot can be a huge waste of time if they are all full. Our product solves this issue by keeping track of the number of people entering and leaving a room. When applied to a study space such as a library, students could use this to see if there will be room for them if they go there. Some features include accurate tracking of single objects, multiple room tracking so that all study spaces can be tracked at once, and a design with no extra steps for people entering a room.

Our system performs well in general, but there are a few scenarios where it does not display great accuracy. First, there is no way for our ultrasonic sensors to detect multiple people entering at the same time as it basically treats them as one object entering the room. Additionally, two people entering in quick succession or one person moving slowly could trigger the sensors wrong because of the rest period resulting in miscounts. Finally, the system cannot differentiate between a person and some other object entering the room. The other problems are still present with an inanimate object moving through the system, so this is the biggest problem of all.

It is difficult to determine the exact frequency of these problems in a real-world situation, but if they are uncommon, then the system could be extremely impactful in solving the problem of wasting students' time. A student would never have to go to a place with no spots since they could see how many available study spaces there are before starting to go there. This device could save \$1.6 billion of students' time per finals season by implementing it all over the world.

Further development of the alpha prototype could include upgrading the ultrasonic sensors to a thermal sensor to track heat signatures. This could fix the problem of

multiple people entering at once and even fix the problem of entering too quickly or slowly as well as handling the problem of detecting non-human objects entering the room.

III. INTRODUCTION

The problem we were aiming to solve was students wasting time trying to find study space during finals week. Students' time is incredibly valuable since they need to study all the can for their future. We identified important parts that our design must include. First, it must be cheap to implement. The school would have to purchase and install these devices in their buildings. If they were too expensive, lots of schools would not have the budget to implement them, defeating the purpose. Second, the design must be non-invasive. If the device is too much of an eyesore, no one would want to install it in their buildings. It has to be fairly small and non-noticeable. Next, it has to create no work for the student. Students are busy and they will not want to interact with a machine when they are just trying to go into a building. Therefore, if the device requires students to give it an input, the device will be too inconsistent since most students would not bother. Lastly, the deice should be easy to use for the consumer. If a student has to jump through many hoops just to get the information from our device, they won't use it.

IV. DESIGN CONSIDERATIONS

The design process began with the problem we were trying to solve. We needed a way to track the people in a study space, and a way to relay that information to a consumer. First we brainstormed about the former. We all wrote down everything we could think of and compared to each other. The mechanisms we thought of were motion detectors, thermal sensors, ultrasonic sensors, cameras, and buttons. We used the same method to brainstorm for the information mechanism. We came up with using an app and a wireless signal, uploading the information to a website, and a physical display Infront of the place itself. We then went through and weighed the pros and cons of each option.

Starting with the tracking mechanism, we started exploring the motion sensor. It would trigger when motion went past it, which would be good for counting the people in the room, however it failed to track direction, causing it to be useless. Then the thermal sensor seemed promising, it would easily be able to differentiate people and anything else passing through a doorway, however after seeing the price of the thermal sensors we could theoretically use, we scrapped that pathway. Cameras seemed like a good option, however it would have to be paired with software that would be able to identify a person and what direction they were going. We thought that we could just place many cameras facing all chairs, but found that to be a serious privacy issue, and that would also make our system no more than a glorified security system, which is not what we wanted. The button would be pressed by a person either entering or leaving, which would track the population of the room well, except there was a glaring issue. That requires work on each and every person coming into a room, which we know is highly improbable that people would actually care enough to engage with the system. The ultrasonic sensors seemed to have few flaws, except for they could not identify direction like motion sensor. We realized we could get around this by having a system of two sensors, and track which or when they are triggered to find direction. We could have used this on either the motion sensor or the ultrasonic sensor. We decided on the ultrasonic sensor for a few reasons. One, we were familiar with how to code with it from project 2, and they were cheap and would make it easy to keep costs down. We had the option of using a second, larger sensor with a longer cable. We wanted to use this, however it made a loud clicking noise while it sensed. That would not be viable for a device used in a study space, so we discarded that idea.

Moving on to the information delivery system. We quickly ruled out the physical display, since it would require someone to already be at the study site, defeating the purpose of the product. A website could be just as effective as an app, however since everyone always has access to a phone, we decided that an app would be easier for the consumer than a website. We did not end up being able to create an app for the alpha prototype. In place, we had a microbit display that shared the information through radio signal. It displayed the same information that would be found in the app, just with lower complexity and visual UI.

Next came the software. We had two main designs to choose between from to track entering and leaving. The first was to have both sensors sense constantly, and will trigger when something comes closer than the base level. Entering or leaving would be determined based on which sensor was triggered first. To prevent multiple triggers happening on one person, there must be amount of sleep time on the device after triggering. The second also had each sensor triggering constantly, but would track the time in between one sensor being triggered and the other, finding direction based on that. Both cases had possible error. The first case had error when multiple people entered too quickly, or when a group of people entered. The second case had the same issues because it still needed to sleep, but it also had a lower success rate of triggering correctly when only one person entered. We went with the first option because of this.

We also had to figure out how we wanted to mount the sensors for the prototype. To increase concealment, we had the two circle parts of the sensors pushed into holes of the same size in cardboard. We determined that for some reason the cardboard touching the metal part of the sensor interfered with it, so we had to scrap that.

V. RESULTS AND DISCUSSION (JENSEN)

The implementation and testing of StudyTrack was very successful, meeting all the engineering specifications outlined in the project proposal. The primary technical objective was to create a system that was accurate, affordable, and non-invasive, and the final prototype satisfied these requirements. Functional testing of the motion sensor integration demonstrated a near 100% accuracy rate in tracking entries and exits during controlled trials. This nearly meets the target and surpasses the threshold of 95%. This level of accuracy is critical for the system to reduce student stress, because it needs to be reliable in order for the system to be effective in saving time.

To continue, the prototype followed constraints necessary for scalability. The final unit cost was kept within the \$15 target, significantly lower than the \$30 maximum threshold. This economic efficiency validates the potential for mass deployment across

campus locations like WALC and Honors Great Hall, without creating a financial burden on the university. Also, the physical footprint of the device was maintained under 0.5 cubic feet, ensuring the hardware remains non-intrusive and does not interfere with the functional environment of library and study spaces.

The implications of these results offer a distinct improvement over the current state-of-the-art. While the existing Microsoft Teams reservation system suffers from "ghost bookings", where rooms remain marked as busy even after students leave, our system provides real-time validity. By requiring only two steps to use (passive entry/exit tracking and a glance at the display), we successfully minimized the user interaction load, achieving our usability target.

Ultimately, the technical success of this device translates directly to the stakeholder benefits identified in our needs analysis. By eliminating the 15 to 60 minutes students typically waste wandering for a seat, students can use that time to study and improve their grades. By solving the specific "pain point" of uncertainty and time loss, the project successfully demonstrated that a low-cost, automated solution can optimize study habits, decrease anxiety during finals week, and improve overall academic productivity.

VI. CONCLUSIONS AND RECOMMENDATIONS (JENSEN)

Although our project worked as intended, there were some minor issues that we found that would take longer to fix and come up with a solution.

One issue that arose was that when multiple people enter a room, there will be a slight miscalculation in the amount of people because there is sleep time in the code, which is a slight issue. Another similar issue is if multiple people enter a room at the same time; the program shows that only one person walked in (because of sleep time), and this makes the program inaccurate. Also, if an object like a backpack came through the machine, it may count it as a person and overestimate the count. One way that we came up with to solve this issue would be to use a thermal sensor to count the number of heat signatures, which would eliminate the problem of an overestimate or underestimate of people, as well as the counting of inanimate objects like notebooks or backpacks.

To continue, in bigger halls like WALC and the Honors Great Hall, there are multiple entrances and exit doors. The problem with this is that this would raise the costs significantly, because there would need to be a sensor near every door, which would also make it seem more invasive and ruin the aesthetic of a study room. To solve this, we would most likely implement a different solution, something more expensive than one sensor, like a camera that has a view of all the doors and can count the people coming in and out efficiently and accurately. This was one of our alternative solutions that we decided not to pursue because of its difficulty and money restrictions.

A question that rose during our presentation was the similarity of our project to the system that Purdue implemented in CoRec. CoRec's system shows the rooms (Upper Gym, Basketball Courts, Lower Gym, etc) and the amount of people in each one and its maximum occupancy followed up with a visual progress bar that also shows the time it last updated. The difference between our project and CoRec's is that our program updates instantly, allowing the student to have accurate information at all times, because students don't want to know how the occupancy from 30 minutes ago, they want to know the current occupancy.

If the project were to be continued, another step we would take is to create an interface for the student to check the occupancy. Currently, we are using a microbit, which has limitations because it cannot be widely used and it can only display so much information. Our interface would be something either on a website or an application that students can download on their cellphones.

To conclude, if we were to improve the project and had more funding, a thermal sensor would definitely be used due to its accuracy and overwhelming benefits over our current system, as well as an interface for the student to use. But, with the tools that we were allocated, our project exceeded our expectations and successfully proved that an affordable, real-time occupancy tracker is a viable solution to the campus study space crisis.